

Xiaotong Shen

+1(647)-404-4163 | xiaotong.shen@mail.utoronto.ca | [linkedin.com/in/shen-xiaotong](https://www.linkedin.com/in/shen-xiaotong) | github.com/xiaotong-shen

Education

University of Toronto

Honours Bachelor of Science (HBS) in Data Science, Computer Science + ASIP Co-op

Major GPA: 4.0/4.0, cGPA 3.75/4.0

Relevant Coursework: Linear Algebra, Multivariable Calculus with Proofs, Software Design, Probability and Statistics I & II, Computer Organization, Machine Learning, Data Structures and Analysis, Data Science

Seeking 4–16 Month

Co-op, 2026

Experience

Research Assistant, Software Developer | Java, Typescript COBWEB

Toronto

August 2025 - Present

- Implemented **XCS** (Accuracy-based Learning Classifier System) within an existing ecological simulation, prioritizing **rule-based interpretability** over black-box RL alternatives to support domain researcher understanding.
- Conducted **iterative user testing** with **15+ high school researchers**, evaluating visualization techniques and simulation behavior through structured experimentation to assess pedagogical effectiveness.
- Performed **literature review** on reinforcement learning and ecological modeling frameworks to inform implementation decisions, producing **technical documentation** for future developers.

Outreach Coordinator

SDSS & UTMIST, University of Toronto

Toronto

2025 – 2026

- Led end-to-end planning of a **Python workshop** for **50+ students**, analyzing historical attendance data to inform beginner-focused programming and timing decisions at the start of the academic year.
- Coordinated on planning **5 events** across SDSS and UTMIST, managing speaker outreach, logistics, and cross-team collaboration to support two of UofT's largest data science and ML communities.
- Managed **sponsor and partner relationships**, conducting cold outreach to companies and speakers and producing marketing content to grow community engagement.

Intern | Python, Excel

Student Success Programs, University of Toronto

Toronto

May 2025 – August 2025

- Collected and analyzed engagement data from **5,000+** student interactions (email open rates, event attendance) with **Python** to identify behavioral trends and recommend improved outreach strategies.
- Authored and presented a comprehensive end-of-term report on content engagement, with **5+ actionable recommendations**, informing future program design, data collection practices, and outreach strategies.
- Designed and published social media content for **@sidneysmithcommons** on Instagram, generating **100,000+ cumulative likes** to increase student awareness of programming.

Section Leader

Stanford University (Code in Place)

Remote

May 2025 – July 2025

- Instructed weekly tutorials for a cohort of **20+** students on core **Python** concepts such as **data structures, functions, and memory management** as a part of **Stanford University's Code in Place** program.

Projects

AURA | Python, PyTorch, HuggingFace

UTMIST (2025–2026)

- Fine-tuned a **binary personality classification model** for emotion recognition in conversational agents, reframing the task from regression to classification based on psycholinguistic research on Big Five trait expression.

VisTeX | TypeScript, Python, LaTeX

Github (2025)

- Built a **VS Code extension** that links LaTeX source, compiled PDF, and interactive **Plotly** visualizations in one workflow, rendering 2D/3D plots (surface, curve, contour, parametric) inline with proof environments.

Web App for Predicting TTC Delays | Javascript, Python

Github (2025)

- Built an **interactive data dashboard** using **Streamlit** and **Plotly** to visualize real time TTC delay predictions, designing charts and reports that surface model outputs in a clear and accessible way.

Technical Skills

- Programming Languages:** TypeScript, Javascript, PostgreSQL, Python, Java, R, SQL
- Frameworks & Tools:** React, Node.js, TensorFlow, PyTorch, Git/GitHub
- Software Development:** Object-Oriented Programming, Data Structures & Algorithms